On the Cohen–Macaulayness of abelian extensions of regular local rings

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Introduction

Notes I made for my talk at BIKES:

> Let R be a regular local ring with quotient field K. Let L be a Galois extension of K and S the integral closure of R in L. Assume that the Galois group is abelian. Is S Cohen–Macaulay? Roberts showed the answer is affirmative when the order of the group is invertible in the residue field of R, and Koh gave a counterexample in mixed characteristic. Recently, in joint work with Singh and Sridhar, we give counterexamples in equicharacteristic. We shall discuss all these results, having rather elementary constructions and proofs!

§1. Nonmodular

Paul Roberts [Rob80] showed the following:

Theorem 1.1. Let R be a regular local ring with fraction field K. Let L be a finite Galois extension of K with an abelian Galois group. Assume moreover that the order of the Galois group is not divisible by the characteristic of the residue field of R. Let S denote the integral closure of R in L. Then S is a Cohen–Macaulay ring.

We observe some generalities. The separability of L/K implies that S is a finite R-module. Thus, the Cohen–Macaulayness of S is equivalent to its R-freeness.[1] Let G denote the Galois group of L/K. The group G acts on L and S while fixing K and R, respectively. Thus, L is a $K[G]$-module and S an $R[G]$-module. Finiteness of $R \subseteq S$ gives us an isomorphism $S \otimes_R K \cong L$, which is also one of $K[G]$-modules. (The isomorphism is saying that every

[1] Use the Auslander–Buchsbaum formula, recalling that R is regular.

element of L can be written as a fraction s/r with $s \in S$ and $r \in R$.) On the other hand, we also have $L \cong K[G]$ in view of the Normal Basis Theorem, giving us

$$K[G] \cong S \otimes_R K. \tag{1.1}$$

We now outline the main ideas of the proof. Roberts first reduces the problem to the case that R is complete with an algebraically closed residue field k; we assume that this is now the case. Writing G as a sum of cyclic groups $\mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$, we see that the group ring $k[G]$ is isomorphic to

$$k[G] = \frac{k[x_1]}{(x_1^{d_1} - 1)} \times \cdots \times \frac{k[x_k]}{(x_k^{d_k} - 1)}.$$

The nonmodularity implies that each d_i is coprime to $\operatorname{char}(k)$. As k is algebraically closed, each factor above splits into linear factors, giving us that

$$k[G] \cong k_1 \times \cdots k_n; \quad \text{each } k_i \text{ isomorphic to } k.$$

As R is complete, the above decomposition be lifted to $R[G]$, giving us

$$R[G] \cong R_1 \times \cdots R_n; \quad \text{each } R_i \text{ isomorphic to } R.$$

As S is an $R[G]$-module, it must decompose as

$$S \cong S_1 \times \cdots S_n,$$

where each S_i is an R-module. Moreover, the above decomposition is *canonical*! The module S_i is given as the elements of S annihilated by elements of $R_1 \times \cdots \times R_n$ whose ith component is zero.

In view of (1.1), we see that $K[G]$ can be decomposed in the following ways:

$$K[G] \cong R[G] \otimes_R K \cong (R_1 \otimes K) \times \cdots \times (R_n \otimes K) \cong K \times \cdots \times K,$$
$$K[G] \cong S \otimes_R K \cong (S_1 \otimes K) \times \cdots \times (S_n \otimes K).$$

As the above decompositions are canonical, we must have $S_i \otimes K \cong K$ for each i. However, S is a reflexive R-module, and thus the same is true for S_i. Thus, each S_i is a reflexive R-module of rank one. As R is a UFD, this implies that each S_i is isomorphic to R. In turn,

$$S \cong S_1 \times \cdots \times S_n \cong R^n \quad \text{as } R\text{-modules.}$$

In fact, factoriality of R was all that we used to conclude that S is free. Thus, we obtain:

Porism 1.2. Let R be a UFD with fraction field K. Let L be a finite Galois extension of K with an abelian Galois group. Assume moreover that the order of the Galois group is not divisible by the characteristic of the residue field of R. Let S denote the integral closure of R in L. Then S is free over R.

If we assume G to only be nilpotent, the theorem fails [Rob80, Example 2].

§2. Mixed characteristic

In the same paper, Roberts [Rob80, Example 1] also showed that the hypothesis on the order cannot be dropped in Porism 1.2:

Example 2.1. Consider the regular ring

$$R \coloneqq \left(\frac{\widehat{\mathbb{Z}}_2[X,Y,U,V]}{(Y^2 - 4U - X^2V)} \right)_{(2,X,Y,U,V)},$$

where $\widehat{\mathbb{Z}}_2$ denote the ring of 2-adic integers. Then, one checks that R is a UFD by showing that 2 is prime and that $R[2^{-1}]$ is a UFD. Letting K denote $\mathrm{Frac}(R)$ and L the Galois extension $K[\sqrt{V}]$, one checks that the integral closure S is not free over R. In fact, Roberts shows that no nonzero S-module can be free over R.

Six years later, Koh [Koh86, Example 2.4] showed that the hypothesis on the order cannot be dropped in Theorem 1.1 cannot be dropped either:

Example 2.2. Pick a regular local ring R such that $\dim R \geqslant 3$ such that $\sqrt{-3}$ is part of a regular system of parameters of R.[2] Pick $x, y \in R$ such that $\sqrt{-3}, x, y$ is a part of a regular system of parameters. Consider $a \coloneqq xy^4 + 27$ and $b \coloneqq x^4y + 27$, and define $\theta_1 \coloneqq \sqrt[3]{ab^2}$ and $\theta_2 \coloneqq \sqrt[3]{a^2b}$. Let S be the integral closure of R in $L = K(\theta_1)$. The extension L/K is Galois of order 3 but S is not free over R.

§3. Equicharacteristic

Forty years after *that*, we have equicharacteristic counterexamples [MSS26, Theorem 2.2]. Recall that if G is a finite subgroup of $\mathrm{GL}_n(k)$, then G acts on the polynomial ring $S \coloneqq k[x_1, \ldots, x_n]$ via degree-preserving k-algebra automorphisms, where the action of $g \in G$ is given by

$$\begin{bmatrix} x_1 & \ldots & x_n \end{bmatrix}^{\mathsf{T}} \mapsto g \begin{bmatrix} x_1 & \ldots & x_n \end{bmatrix}^{\mathsf{T}}.$$

Note the action of G extends to $L \coloneqq \mathrm{Frac}(S)$ and we have $L^G = \mathrm{Frac}(R) =: K$, the extension L/K has Galois group G, and S is the integral closure of R in L.

Example 3.1. Consider the action of $G = \left\langle \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix} \right\rangle$ on $S \coloneqq \mathbb{F}_p[x,y]$. The generator maps $x \mapsto x + y$ and $y \mapsto y$, using which we see that $R \coloneqq \mathbb{F}_p[x^p - xy^{p-1}, y] \subseteq S^G$. One checks that S is integral over R and $[\mathrm{Frac}(S) : \mathrm{Frac}(R)] = p$ and that R is a polynomial ring (hence normal). Using this, one concludes that $S^G = R$, a polynomial ring.

We recall that an element $g \in G$ is a bireflection if $\mathrm{rank}(g - 1) \leqslant 2$. We have the following

useful theorem of Kemper [Kem99, Theorem 2.1]:

Theorem 3.2. Let K be a field of characteristic $p > 0$, and H a finite subgroup of $\mathrm{GL}_n(K)$, with its natural action on the polynomial ring $T := K[x_1, \ldots, x_n]$. If H is a p-group, and T^H is Cohen-Macaulay, then H is generated by bireflections.

We now construct our example in positive characteristic. Consider the ring

$$T := \mathbb{F}_p[x_1, y_1, x_2, y_2, x_3, y_3]$$

with the natural action of the group $G \leqslant \mathrm{GL}_6(\mathbb{F}_p)$ consisting of matrices of the form

$$\left[\begin{array}{cc|cc|cc} 1 & a & & & & \\ & 1 & & & & \\ \hline & & 1 & b & & \\ & & & 1 & & \\ \hline & & & & 1 & c \\ & & & & & 1 \end{array}\right] \quad \text{for } a, b, c \in \mathbb{F}_p.$$

In other words, we have taken three copies of Example 3.1. Within G, consider the subgroup generated by

$$\left[\begin{array}{cc|cc|cc} 1 & 1 & & & & \\ & 1 & & & & \\ \hline & & 1 & 1 & & \\ & & & 1 & & \\ \hline & & & & 1 & 1 \\ & & & & & 1 \end{array}\right],$$

i.e., the diagonal copy of group from Example 3.1. Then, by Kemper's theorem, the ring of invariants T^H is not Cohen–Macaulay, the only bireflection in H being the identity. On the other hand, the ring T^G is a polynomial ring, stemming from Example 3.1.

Thus, we have a obtained an extension

$$R := T^G \subseteq T^H =: S$$

of rings where the smaller ring is regular but the larger ring is not Cohen–Macaulay. As G was abelian, Galois theory tells us that $\mathrm{Frac}(S)/\mathrm{Frac}(R)$ is Galois with abelian Galois group $G/H \cong \mathbb{Z}/p \times \mathbb{Z}/p$. One may also note that $S \subseteq T$ is Galois (at the field level) with Galois group $\mathbb{Z}/p$ with S being a UFD but T not S-free. The rings above are constructed in the $\mathbb{N}$-graded realm, rather than the local one, but one may localise at the homogeneous maximal ideals to get the desired results.

References

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